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Author(s): William B. English

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Understanding the Costs of Sovereign Default: American State Debts in the 1840's

By WILLIAM B. ENGLISH*

The defaults of U.S. States in the 1840's provide a powerful test of competing models of sovereign debt. The Eleventh Amendment to the United States Constitution prevented foreign creditors from obtaining payment in the federal courts. Moreover, because the defaulting states were part of a large and economically integrated nation, creditors could not enforce payment by imposing military or trade sanctions. In spite of the lack of sanctions, however, most states eventually repaid in full. It appears that the states repaid in order to maintain access to capital markets, much as in reputational models of sovereign debt. (JEL F34)

Jonathan Eaton and Mark Gersovitz (1981) argue that sovereign debtors repay because if they did not they would get a reputation for defaulting and thereby lose access to international capital markets. In contrast, Jeremy Bulow and Kenneth Rogoff (1989a) argue on theoretical grounds that reputational effects are not sufficient to induce countries to repay debts. They argue that foreign creditors must be able to impose direct sanctions in order to induce sovereign debtors to repay. They suggest that trade embargoes may serve such a purpose (see also Bulow and Rogoff [1989b] especially the appendix).

The defaults by U.S. states in the 1840's provide strong evidence that direct sanctions are not needed to induce repayments of sovereign debts. Between 1841 and 1843 eight states and one territory defaulted on their obligations, and by the end of the decade four

states and one territory had repudiated all or part of their debts. These debts are properly seen as sovereign debts both because the United States Constitution precludes suits against states to enforce the payment of debts, and because most of the state debts were held by residents of other states and other countries (primarily Britain). The U.S. states, however, were insulated from direct sanctions that could have been imposed on individual countries because they were part of a powerful union of states. Going to war to recover the debts would have been very expensive given the relative wealth and power of the United States at the time. Cutting off trade with an individual state was difficult because, with free trade between the states, goods could be exported through another state. Cutting off trade with the entire United States in order to retaliate against a defaulting state would have been very expensive for the British, however, since the United States provided a significant market for British exports. Moreover, the most important U.S. export was cotton—the raw material input into one of the most important British industries.

In spite of the inability of the foreign creditors to impose direct sanctions, most U.S. states repaid their debts. It appears that states repaid in order to maintain their access to international capital markets, much like in reputational models. The states that repaid were able to borrow more in the years leading up to the Civil War, while those that did not repay

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The analysis and conclusions set forth in this paper are mine alone, and do not indicate concurrence by the Board of Governors of the Federal Reserve System, its staff, or the Federal Reserve Banks.

were, for the most part, unable to do so. States that defaulted temporarily were able to regain access to the credit market by settling their old debts. More surprisingly, two states that repudiated a part of their debt were able to regain access to capital markets after servicing the remainder of their debt for a time. Thus these cases provide evidence for a signaling model of sovereign debt similar to that presented by Harold L. Cole et al. (1995).

Of course, the fact that U.S. states do not default on their debts today could also be taken as evidence against the sanctions view, since it is still not clear that bondholders could impose sanctions on defaulting states. The 1840's defaults provide more compelling evidence, however, because we can observe that no direct sanctions were imposed following actual defaults and repudiations, whereas one can only conjecture what the costs would turn out to be today. Indeed, there is reason to believe that sanctions would be easier to impose today than they were in the 1840's. The fraction of state debts held outside the issuing state, either in other states or abroad—which was very large in the 1840's—is likely smaller now, owing, in part, to the favorable federal and state tax treatment of state securities. Defaults on state residents, however, might lead to adverse political consequences for the politicians supporting default, while defaulting on residents of other states might lead to intervention by the federal government. Moreover, the far greater relative size of the federal government today provides much larger scope for such intervention than was the case in the pre-Civil War era. Even in the case of foreign bondholders, the federal courts' interpretation of sovereign immunity has been narrowed in some respects since World War II, and so legal action might be more effective than it would have been a century and a half ago (see the discussion in Lewis S. Alexander [1987]).

I. State Debts as Sovereign Debts

A. Legal and Constitutional Issues

One might expect that U.S. states could be compelled to repay debts by the federal courts. The power of the federal courts to do so was tested in *Chisholm v. Georgia* (1793), in

which a citizen of South Carolina sued the state of Georgia for nonpayment of a debt. The first Supreme Court decided that it had jurisdiction in such cases, and it found against the state (John V. Orth, 1987 Chapter 2). In response to this decision, Congress passed the Eleventh Amendment to the Constitution, and within two years it had been passed by the required two thirds of the states. The amendment declares that "The judicial power of the United States shall not be construed to extend to any suit in law or equity, commenced or prosecuted against one of the United States by citizens of another state, or by citizens or subjects of any foreign state." This amendment reversed *Chisholm*, and made it very difficult for creditors to force states to repay debts.

An obvious route around the Eleventh-Amendment obstacle is to arrange to have the suit brought by an agent not covered by the amendment. There are four possibilities: a citizen of the state, the government of another country, the federal government, or the government of another state. The first two possibilities have been ruled out by Supreme-Court decisions. In *Hans v. Louisiana* (1890) the Supreme Court ruled that the federal courts do not have jurisdiction over suits brought against a state by a resident of that state (Orth, 1987 pp. 140–41). In *Monaco v. Mississippi* (1934) the court found that, since foreign states are immune from suits by U.S. states, U.S. states should be immune from suits by foreign countries (Orth, p. 141).

In contrast, the Supreme Court has been willing to accept jurisdiction in suits between the federal government and a state and between two states. In *United States v. North Carolina* (1890) the Supreme Court ruled that it had jurisdiction over a suit by the federal government for payments that the government claimed were owed by North Carolina. (The court, however, found for the state.) In 1904 a narrowly-divided court sided with South Dakota in a suit it brought to obtain payment on Reconstruction-era North Carolina bonds that had been donated to the state. While this method of forcing states to pay their debts worked, other states were unwilling to bring such suits, and South Dakota refused a second gift of bonds (Orth, 1987 pp. 83–85).

Another possible route around the Eleventh Amendment is to turn to the state courts. This route seems unpromising, given the likely unpopularity of the bondholders, but it was tried in two states following the defaults in the 1840's. In Mississippi the bondholders won a ruling by the state Supreme Court, but the state constitution did not provide for any way for the judgment to be collected (Orth, 1987 p. 45). Thus the bondholders were able to show the justice of their claims, but were unable to force the state to pay. Similarly, holders of defaulted Arkansas bonds were able to obtain judgments in that state's Supreme Court declaring that their claims were proper, but they too were unable to collect on the bonds (Reginald C. McGrane, 1935 pp. 258–59, 263–64).

B. *Who Were the Bondholders?*

The defaulted state debts in the 1840's were not generally owed to state residents. Most of the state bonds were sold in Europe (primarily London), New York, or Philadelphia. For example, the par value of the bonds of Pennsylvania—the most heavily indebted state—was \$34.5 million in 1842. Of this, over \$20 million was held in England, and an additional \$1.8 million in Holland. Even France, which held \$570,000 of Pennsylvania bonds, held more than any U.S. state other than Pennsylvania itself (McGrane, 1935 p. 71n).

Although detailed information is not available for other states, a substantial fraction of many states' bonds were held abroad. For example, about two fifths of the New York debt was held in Europe in 1843, and some of the bonds held in this country may have been held by agents of foreign investors (Thomas P. Kettell, 1848 p. 252). About half of the Florida and Arkansas bonds were held in Amsterdam or London (McGrane, 1935 pp. 228–29, 251–52). By 1846 a substantial majority of the Illinois canal debt was held in London (F. Cyril James, 1938 v. 1, p. 169). The governor of Mississippi claimed that all of that state's debt was held by foreigners in January of 1840 (Robert Lowry and William H. McCardle, 1891 p. 289). Some states even made their bonds payable in foreign currencies in European cities in order to sell them more easily in

those markets. For example, nearly half of the Maryland debt was denominated in pounds sterling and was payable in London (U.S. Congress, House of Representatives, 1843b Report No. 296, pp. 56–57; Kettell, 1849b p. 491). Similarly, over two thirds of the Georgia debt was denominated in pounds and payable in London, as was over 70 percent of the debt of Mississippi (U.S. Congress, House of Representatives, pp. 68–69; McGrane, 1935 p. 197) and more than 90 percent of the debt of Louisiana (Benjamin R. Curtis, 1844 p. 137; McGrane, pp. 171–75).

Even if bonds were held in the United States, the holders could still be residents of other states. For the large northeastern states, bonds not held abroad were mostly held in the state. Less than 5 percent of the New York and Pennsylvania bonds were held in other states (McGrane, 1935 p. 71n; Kettell, 1848 p. 252). In contrast, the Ohio debt was explicitly divided into a “domestic” debt—payable locally—and a “foreign” debt which was payable in New York. The foreign debt was about three quarters of the total (U.S. Congress, House of Representatives, pp. 82–85). Similarly, all of the Indiana bonded debt was “foreign debt” (Logan Esarey, 1918 v. 1, p. 435).

II. State Debts in the 1830's and 1840's

Between 1820 and 1839 the debts of U.S. states increased by a factor of thirteen. As can be seen in Table 1, these debts were primarily for two purposes: transportation improvements and banking. The success of the Erie Canal had shown that state investment in internal improvements could be productive and profitable. Other states, primarily those in the North, attempted to emulate New York's success by building their own canals and railroads. In contrast, southern states borrowed primarily to obtain the capital for banks. The southern states felt that their banking systems were insufficient, especially after the United States Bank was not rechartered. (See McGrane, 1935 Chapter 1.)

A. *Why Did States Default?*

By the fall of 1841, nineteen U.S. states and two territories had issued bonds, while

TABLE 1—STATE DEBTS IN SEPTEMBER 1841
(THOUSANDS OF DOLLARS, BY PURPOSE OF ISSUE)

State or territory	Purpose			Total debt
	Transportation	Banking	Other or not available	
Alabama	0	15,400	0	15,400
Arkansas	0	3,176	0	3,176
Connecticut	0	0	0	0
Delaware	0	0	0	0
Florida Territory	0	3,900	100	4,000
Georgia	1,321	0	0	1,321
Illinois	9,517	3,035	975	13,527
Indiana	8,998	2,390	2,863	14,251
Iowa Territory	0	0	0	0
Kentucky	3,046	0	40	3,086
Louisiana	1,200	17,389	585	19,174
Maine	0	0	1,735	1,735
Maryland	11,789	0	226	12,015
Massachusetts	5,675	0	294	5,969
Michigan	5,320	0	291	5,611
Mississippi	0	7,000	0	7,000
Missouri	20	389	433	842
New Hampshire	0	0	0	0
New Jersey	0	0	0	0
New York	20,822	0	975	21,797
North Carolina	0	0	0	0
Ohio	13,594	0	1,489	15,083
Pennsylvania	30,024	0	6,342	36,366
Rhode Island	0	0	0	0
South Carolina	3,350	138	2,203	5,691
Tennessee	1,716	1,500	0	3,216
Vermont	0	0	0	0
Virginia	6,193	801	0	6,995
Wisconsin Territory	0	0	418	418

Source: See the Data Appendix.

seven other states and one territory (Iowa) did not have any debt. Of the borrowing states, eight would default within two years: Arkansas, Illinois, Indiana, Louisiana, Maryland, Michigan, Mississippi, and Pennsylvania. In addition, a territory (Florida) repudiated its debt.¹

Although the borrowing was for different causes, the basic problem was the same for

states in the North and the South. The long inflationary boom of 1834–39, which had been checked by the panic of 1837, came to an abrupt end in 1839. The causes of the collapse are not entirely clear, but probably involved a tightening of credit by the Bank of England and a resulting outflow of gold from the United States to Britain (Peter Temin, 1969). In any case, the transportation investments of many northern states were not yet complete, and additional credit was not available. Since the incomplete projects generated little revenue, the bonds issued for their construction became a burden for the states. In the South, many of the banks that had been started with the borrowed funds failed in the few years after 1839, and the bonds issued on their behalf fell on state budgets.

¹ In addition, the Territory of Wisconsin seems to have repudiated a portion of its debts (U.S. Congress, House of Representatives, 1843b, Report No. 296, pp. 48–49). Given that the repudiation seems likely to have resulted from malfeasance on the part of the state agent, rather than the territory's unwillingness or inability to pay, Wisconsin is not counted in this paper as a defaulting or repudiating state.

The troubled states (i.e. those that defaulted or repudiated) were considerably more indebted than the other states.² As can be seen in Table 2, the defaulting states had debts of more than 20 percent of annual state income, while those that had debts but did not default had borrowed only about 9 percent of state income on average. Surprisingly, the states that repudiated completely had somewhat lower debt burdens than the states that repudiated partially. In addition, some heavily indebted states did not default (e.g., Alabama) while some states with relatively light burdens repudiated (e.g., Mississippi).

The debt burdens shown in Table 2 are based on debt data from September of 1841 and income data for 1839. Since most states were not able to borrow after 1839, the debt burdens in Table 2 are roughly correct for 1839. As measured by the Warren and Pearson wholesale price index, prices fell by one third between 1839 and 1843, thereby boosting the real burdens of the debts substantially.³ In addition, there may have been a decline in real output between 1839 and 1843, although the

size of this decline is subject to debate (see Temin [1969] and Charles W. Calomiris and Christopher Hanes [1994] for discussions).

B. *Histories of the Defaults and Resumptions*

The defaulting states can usefully be divided into three groups both by region and by the source of their difficulties. Pennsylvania and Maryland, both of which had borrowed to build canals, defaulted temporarily in the early 1840's. Both states eventually paid in full. Three states in the Midwest—Indiana, Illinois, and Michigan—borrowed to build transportation networks. Their difficulties were more serious than those of Pennsylvania and Maryland, and the outcomes for their bondholders were less satisfactory. Finally, three southern states and the Florida Territory issued bonds on the behalf of banks or guaranteed the bonds of banks. When the banks failed in the early 1840's, all four refused to pay all or part of their bonds. (See Table 3.)

*Pennsylvania and Maryland.*⁴—Both of these states borrowed heavily to construct canal systems linking their ports to the Midwest. Although they borrowed large sums, both states were populous and relatively rich, and their debt burdens were lower than those of several other states (see Table 2). In the recession of the early 1840's however, both were temporarily unable to make their interest payments. Although the defaults generated outrage among European investors, neither state was likely to repudiate. After several attempts both states were able to raise their taxes sufficiently to resume the payments owed on their debts, and both states funded their arrears into new bonds.

*Midwestern States.*⁵—The midwestern states of Ohio, Indiana, Illinois, and Michigan all borrowed heavily to build transportation

² The grouping of states by their default status is somewhat problematical. Those that paid without interruption are straightforward, as are those that repudiated more or less completely. The distinction between defaulting without repudiating and partially repudiating is less clear. Pennsylvania and Maryland defaulted, but eventually paid in full (including interest on their arrears). Indiana and Illinois reached settlements that were not quite complete repayments because their arrears were not repaid for several years, and they did not pay interest on their arrears. In addition, the Indiana settlement was ultimately a disaster for the bondholders. Nonetheless, I count these four states as defaulting but not repudiating. Michigan clearly repudiated a part of its debt. Arkansas defaulted on its debts until after the Civil War, but then paid some of its pre-war debts. It is counted here as a partially repudiating state. Louisiana repudiated its bank debt, but paid its remaining debt. In addition, the banks ultimately repaid much of the bank debt. The fraction of debts paid (by the banks as well as the state) may have been quite high, but I still count Louisiana as a partially repudiating state because the eventual outcome was not known in the early 1840's.

³ According to Robert E. Gallman (1964) the GNP deflator fell by only 4.4 percent between 1839 and 1844. Thus the Warren-Pearson index may overstate the decline in prices. Since most out-of-state sales were probably wholesale commodities, the focus here on the Warren-Pearson index seems warranted.

⁴ See Kettell (1847a, 1849a, 1849b) and McGrane (1935 Chapters 4 and 5).

⁵ See Kettell (1849c, 1849d, 1850, 1852) and McGrane (1935 Chapters 6–8).

TABLE 2—DEBT BURDENS

State or territory	Per capita (in dollars)			Ratios (in percent)	
	Gross state product	State debt	Interest on state debt	Debt to income	Interest to income
Those without debt:					
Connecticut	126.67				
Delaware	94.66				
Iowa Territory	52.90				
New Hampshire	89.09				
New Jersey	115.54				
North Carolina	70.99				
Rhode Island	164.26				
Vermont	90.48				
Average:	95.94				
Those that did not default:					
Alabama	73.78	26.06	1.39	35.32	1.88
Georgia	79.35	1.91	0.10	2.41	0.13
Kentucky	72.39	3.96	0.23	5.46	0.32
Maine	79.35	3.46	0.20	4.36	0.26
Massachusetts	148.95	8.09	0.40	5.43	0.27
Missouri	73.78	2.19	0.16	2.97	0.21
New York	111.36	8.97	0.46	8.06	0.41
Ohio	66.82	9.92	0.59	14.85	0.89
South Carolina	77.95	9.58	0.53	12.29	0.68
Tennessee	65.43	3.88	0.21	5.93	0.32
Virginia	75.17	5.64	0.33	7.50	0.44
Wisconsin	111.36	13.49	0.81	12.11	0.73
Average:	87.28	7.88	0.43	9.03	0.49
Those that defaulted temporarily:					
Illinois	65.43	28.42	1.71	43.44	2.61
Indiana	57.07	20.77	1.07	36.40	1.87
Maryland	87.70	25.56	1.39	29.15	1.58
Pennsylvania	104.40	21.09	1.05	20.20	1.01
Average:	86.86	22.69	1.19	26.13	1.37
Those that partially repudiated:					
Arkansas	94.66	32.41	1.93	34.24	2.04
Louisiana	157.30	54.47	2.74	34.63	1.74
Michigan	61.25	26.47	1.59	43.21	2.60
Average:	117.27	42.24	2.25	36.02	1.92
Those that repudiated completely:					
Florida Territory	96.05	74.07	4.41	77.12	4.59
Mississippi	116.93	18.62	0.98	15.92	0.84
Average:	114.31	25.58	1.41	22.38	1.24
All states and territories:					
Average:	90.48	11.79	0.63	13.03	0.70

Notes: Debt and interest data are as of September 1841. State income data are for 1839. For definitions of categories, see text note 2. Averages are weighted by income.

Sources: See the Data Appendix.

networks. Of the four, only Ohio was able to avoid default.

The Illinois and Indiana debts were accumulated while the states constructed major ca-

nal networks linking their states with the Great Lakes and, via the Erie Canal, the eastern seaboard. In the early 1840's both states were unable to borrow to complete their canals and

TABLE 3—DATES OF DEFAULTS, REPUDIATIONS, AND RESUMPTIONS

State or territory	Default	Resumption/repudiation	Outcome
Those that defaulted temporarily:			
Illinois	January 1842	Resumption in July 1846.	Virtually complete repayment.
Indiana	July 1841	Resumption in July 1847.	Debt split (see text). State debt: virtually complete repayment. Canal debt: not clear, but the canal ultimately failed.
Maryland	January 1842	Resumption in January 1848.	Virtually complete repayment.
Pennsylvania	August 1842	Resumption in February 1845.	Virtually complete repayment.
Those that partially repudiated:			
Arkansas	July 1841	Resumption in July 1869.	Low, since no interest was paid on the arrears. The Holford bonds were repudiated in 1884 (see text).
Louisiana	February 1843	Debt split (see text). State debt proper: resumption c.1844. Property bank bonds: not paid by the state.	Virtually complete repayment. The bonds repaid by the property banks. The amount repaid is not clear, but was probably fairly high in some cases.
Michigan	July 1841	Debt split (see text). Fully paid bonds: resumption in January 1846. Part paid bonds: resumption in July 1849.	Virtually complete repayment. About 30 cents on the dollar.
Those that repudiated completely:			
Florida Territory	January 1841	Repudiated in February 1842.	Florida made no further payments on the bonds. Some portion of the bonds were paid by the companies for which they had been issued. Total payments were likely small.
Mississippi: Planters Bank Bonds	March 1841	Repudiated in November 1852.	Mississippi made no further payments on the bonds. Between 1848 and 1852 the accumulated fund was distributed to the bondholders. An additional \$54,000 of these bonds were received by the state in payment for public lands in the 1850's.
Union Bank Bonds	May 1841	Repudiated in February 1842.	Mississippi made no further payments on the bonds.

Sources: Illinois: J. W. Putnam (1909), *Laws of the State of Illinois* (1845); Indiana: Kettell (1849c); Maryland: Kettell (1849b); Pennsylvania: Kettell (1849a); Arkansas: McGrane (1935), Scott (1893); Louisiana: McGrane (1935); Michigan: Kettell (1850); Florida: McGrane (1935); Mississippi: Corporation of Foreign Bondholders (1932).

unable to pay the interest due on their bonds without the canals, hence both defaulted on their interest payments. Ultimately, both states reached agreements with their creditors under which the creditors provided additional funds with which to complete the main canals (the Illinois and Michigan Canal in Illinois and the Wabash and Erie Canal in Indiana). In return,

the states resumed interest payments on their bonds. In addition Illinois deeded the Illinois and Michigan canal in trust to its creditors, and Indiana agreed to pay half the bonds and the Wabash and Erie canal was held in a trust to pay the other half.

Michigan followed a pattern similar to Illinois and Indiana, borrowing heavily to

build a transportation network, although Michigan was building railroads, not canals. As in the other two states, the investments were not completed, and the state was unable to pay the interest on the debts accumulated for them. In addition, the state lost a great deal of money in its dealings with the Morris Canal and Banking Company. It sold almost all of its bonds to the company in return for a series of future payments. When the company failed in 1841 the state had received only \$2.15 million in return for the \$5 million in bonds turned over to the company. The state went into default in 1841, and was unable to obtain a final settlement with its bondholders until 1848. Under the terms of the settlement, the state repaid only the fraction of the debt for which it had received payment from the Morris Canal and Banking Company plus accumulated arrears.

*Southern States.*⁶—Most of the debts of the southern states resulted from issuing state bonds to banks for them to sell to raise capital, or from guaranteeing the bonds of banks in order to aid them in obtaining capital. Many states assisted banks in this way in the 1820's and 1830's (see Table 1).

The state of Louisiana was the third most heavily indebted state in 1841, after Pennsylvania and New York. Its debts had been accumulated primarily to provide the capital for three banks. In 1842 two of the three banks were put into liquidation, and the legislature made no effort to pay the coupons on the state bonds issued for them. The governor claimed that the shareholders of the banks should be forced to pay before the state was required to step in (McGrane, 1935 p. 181).

In 1844 the state legislature divided the state debt into two pieces. The bonds issued directly by the state were called the "state debt proper." These debts amounted to somewhat less than \$4 million, and the state accepted responsibility for them. Most of the remainder of the debt, about \$17.5 million, had been issued for the banks, and the state refused to take responsibility for it. By the late 1840's, how-

ever, two of the three banks had settled their debts by paying off the old bonds or by issuing new bonds backed only by the bank (George Green, 1972 p. 27). The third bank was taken out of liquidation in 1852, and it was expected to raise sufficient money from its shareholders to retire the state bonds (Green, p. 132; see also Ralph Hidy, 1949 pp. 333–35).

During the 1830's Arkansas issued bonds to raise money for the establishment of two banks. The total amount raised for these institutions was over \$3 million—a large amount for so small a state (see Table 2). The banks opened for business in 1838, and suspended in the panic of 1839. In 1840 one of the banks hypothecated an additional \$500,000 of bonds with a bank, which then transferred them to a London investor, James Holford. This transaction was in violation of at least the spirit of the state law authorizing the bonds, which required the sale of the bonds at par.

In 1841 both banks defaulted on the bonds. As in the case of Louisiana, the state refused to make the payments on the state bonds issued for the banks, and the bondholders found it difficult to collect from the banks. Although the Arkansas Supreme Court found that the state was liable for the bonds (including the Holford Bonds) if the banks could not pay, the bonds were not settled in the time before the Civil War.⁷

Holders of the Mississippi bonds fared even worse than the holders of the Arkansas bonds. Mississippi issued \$2 million of state bonds to provide funds for the Planters' Bank of Mississippi in 1831–33. In 1837 the state issued another \$5 million of bonds for the Union Bank. The Union Bank bonds were not issued and sold strictly in accordance with state law—violations which later provided a pretext for their repudiation by the state (McGrane, 1935 pp. 197–200).

The banks failed to make interest payments on their bonds in 1841, and the state refused

⁶ See McGrane (1935 Chapters 9–12, 14).

⁷ Several years after the Civil War, in 1869, the Reconstruction government issued bonds paying 6 percent to cover the principal and unpaid coupons on the pre-war bonds. Interest was not paid on the arrears (English, 1991 p. 17n). In 1884, however, an amendment to the state constitution specifically repudiated the Holford bonds (McGrane, 1935 pp. 296–98).

to provide the money. The state repudiated the Union Bank bonds in 1842, but did not repudiate the Planters' Bank bonds. In 1844, the Governor stated that the Planters' Bank bonds were obligations of the state, and should be paid (Governor's message of January 1, 1844, reported in U.S. Congress, House of Representatives [1846 Document No. 226, p. 886]). Nonetheless, the state made no effort to pay the debts. In 1848 the state used the sinking fund of the Planters' Bank to pay some coupons of the Planters' Bank bonds, and allowed holders of these bonds to use them to purchase state lands. Finally, in 1852 the state passed a law by plebiscite repudiating the Planters' Bank bonds.

During the 1830's the Territory of Florida issued \$3 million of bonds for the Union Bank of Florida, and guaranteed \$500,000 of the bonds of the Bank of Pensacola and \$400,000 of the bonds of the Southern Life Insurance and Trust Co. The total amount of debt was not particularly large, but Florida was poor and lightly populated at the time, and the burden of the debt was higher than that of any state (see Table 2).

The Florida banks performed poorly after the panic of 1839. As in Mississippi, the way in which some of the bank bonds were sold provided a pretext for their repudiation. In 1841 the Governor of the territory refused an appeal by the bondholders to pay the interest on the bonds, and stated that the creditors should collect from the banks' stockholders. By January 1844 the Southern Life Insurance and Trust Co. had retired all of its state-guaranteed bonds. The Union Bank also retired a small portion of the state bonds issued for it. (See the annual message of Governor Call in January of 1844, reported in U.S. Congress, House of Representatives [1846 Document No. 226, pp. 776–77].)

III. Implications for Models of Sovereign Debt

There is substantial debate in the academic literature over why a sovereign debtor might choose to repay its debts. Because it is sovereign, it cannot be forced by a court to do so. The economics literature provides two possible explanations for repayment. First, in reputational models repayment safeguards

the borrower's reputation as a "good" borrower, and allows the country continued access to international credit markets (Eaton and Gersovitz, 1981). If the value of being able to borrow in the future is large enough, then the borrower may choose to repay even though doing so is not in its short-run interest.

Bulow and Rogoff (1989a) provide a disarmingly simple counter-argument to the Eaton and Gersovitz model. Their argument can be seen most clearly in a nonstochastic example. Consider a country with high income in even periods and low income in odd periods. For simplicity assume that there is no growth, and that the world real interest rate is fixed and equal to the country's rate of time preference. Eaton and Gersovitz show that there can be an equilibrium in which the country borrows each odd period and repays the loan each even period. In such an equilibrium, the country chooses to repay because the future cost of not doing so—i.e. unsmoothed consumption—is higher than the short-term benefit of higher consumption.

Bulow and Rogoff point out that the country could default in an even period and take the funds that would have been used to repay the creditor and use them instead to smooth consumption from that period on by saving at the world interest rate in even periods and dissaving in odd periods. Doing so makes the country strictly better off because it earns interest rather than pays interest, raising consumption in every future period. Bulow and Rogoff show that this intuitive result holds in a stochastic model as long as actuarially fair payment-in-advance insurance contracts are available. Their result shows that the implicit assumption in Eaton and Gersovitz is that creditors can not only cut off new loans to a defaulting country, but can also interfere with national saving.

Given this theoretical result, and the fact that sovereign debts are often repaid, Bulow and Rogoff conclude that creditors must be able to impose some direct sanction on defaulting sovereign borrowers. Borrowers repay only if the cost of the sanctions is larger than the amount owed. The sanctions could include war, restrictions on international trade, or reductions in international aid.

A. Evidence Against Sanctions

The U.S. state defaults of the 1840's provide strong evidence against the direct sanctions theory of sovereign debts. Some in the United States, including ex-President John Quincy Adams, feared that defaulting on debts owed to British citizens could lead to war with Great Britain (McGrane, 1935 pp. 35, 57). After the defaults had taken place, however, it was clear that this was not the case. The British government refused to get involved in the negotiations over the debts. The British Foreign Secretary, Lord Aberdeen, noted that the bonds were not obligations of the federal government, and so the British government had "no concern with the securities ... and no power to compel payment of the sums required" (quoted in McGrane, p. 53). Similarly, Lord Palmerston, a later British Foreign Secretary, noted that, "British subjects who buy foreign securities do so at their own risk and must abide the consequences" (quoted in McGrane, p. 202).

While British banking houses might have been able impose costs on defaulting states by cutting off trade credit, doing so would have been difficult because of the freedom of trade between states. In addition, it is not clear that British banks had the incentive to impose costly penalties on the states. The banks themselves did not generally hold the state bonds. Instead, the bonds were sold to private investors in Britain (Leland H. Jenks, 1927 p. 79). In any case, the available data on trade, import prices, and export prices show no indication of such sanctions. (See English [1991 pp. 22–23], for a discussion.)

Similarly, there is no evidence that bondholders in the United States were able to impose sanctions on the defaulting states. One obvious possibility is that bondholders who were citizens of the defaulting state could have attempted to defeat the politicians who voted for default when they were up for reelection. In practice, however, this does not seem to have been done, although foreign bondholders did attempt to lobby elected officials in some states in order to gain support for resumption (McGrane, 1935 Chapter 3). In many cases, however, default, and even repudiation, were quite popular with the electorate—perhaps be-

cause of the low level of debts held in state and, more speculatively, the concentration of the domestic holdings. (See English [1991] for a discussion of the politics of default.)

Given the evident lack of direct sanctions, models of sovereign debts based on such costs suggest that all of the states should have repudiated. As noted above, however, all of the states except Mississippi and Florida eventually paid at least part of their debts. One could claim that the states that paid were threatened with sanctions and so chose to pay, and the states that did not pay were the ones on which no direct sanctions could be imposed. Thus no sanctions were imposed, but they still played an important role in inducing the repayments actually made. But this argument is very weak: why could costs be imposed on Alabama but not Mississippi, on Illinois and Indiana but not Michigan?

B. The Costs of Default

Rather than direct sanctions, the cost of default appears to have been loss of access to new loans. The loss of access, in turn, appears to have been the result of damage to defaulting states' reputations in credit markets. The states that serviced their debts were able to borrow again in the 1840's and 50's while those that repudiated found it difficult to do so. Of the eleven states that repaid without interruption in the early 1840's, all were able to borrow (U.S. Department of Interior, 1884 v. 7, pp. 523–639). Indeed, all but three of these states (Maine, South Carolina, and Alabama) had larger debts in 1860 than in 1841. Maine and South Carolina paid off a substantial fraction of their debts in the 1840's, but borrowed again in the 1850's. Alabama, which had the highest debt burden of the states that did not default, did not increase its debt but did issue new bonds in order to retire old bonds.

In contrast, the states that repudiated all of their debts did not borrow significant sums in the period before the Civil War. Neither Mississippi or Florida issued new bonds, although both states borrowed small amounts, presumably by issuing warrants or by borrowing from state trust funds (U.S. Department of Interior, 1884 v. 7, p. 588). The Barings tried to induce Mississippi to resume payments on its debts

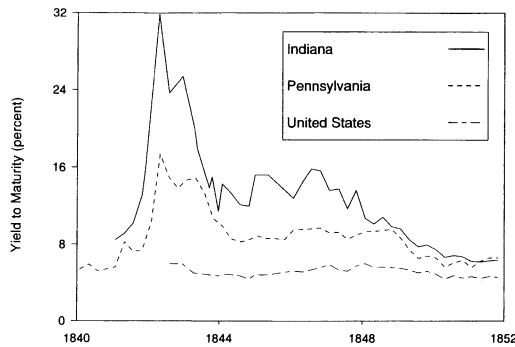


FIGURE 1. UNITED STATES, PENNSYLVANIA, AND INDIANA BOND YIELDS, 1840–1852

when the state was trying to raise money for the Mobile and Ohio railroad, but this effort failed (Hidy, 1949 p. 336). Florida decided to issue bonds for railroads in the 1850's, but it appears that the state did not actually issue any bonds until the 1860's (U.S. Department of Interior, v. 7, pp. 587–89; Carter Goodrich, 1960 p. 160).

The experiences of the states that partially defaulted were more varied. As noted above, in the period before the Civil War Arkansas did not take responsibility for the bank bonds it had guaranteed. As a result, it looked much like Mississippi and Florida, and, like those states, it did not borrow again until after the war. In the 1850's some in the state argued that the defaulted debts should be settled in order to allow the state to borrow to build railroads, but this was not done (McGrane, 1935 p. 262).

Surprisingly, the other two states that partially repudiated their debts were able to regain access to credit markets by repaying only a fraction of their debts. As noted above, Louisiana paid its state debt proper while not delivering on its guarantee of bank bonds. Nonetheless, by the mid-1850's yields on Louisiana bonds were similar to those on the bonds of other states—such as Georgia, North Carolina, Tennessee, and Virginia—which had not defaulted (Benjamin U. Ratchford, 1941 p. 133). In 1854 the state borrowed more than \$2.5 million (U.S. Department of Interior, 1884 v. 7, p. 597).

One could argue that the Louisiana default would have been more costly except for the ability of the Louisiana banks ultimately to pay back much of the debt that had been guaranteed by the state. This argument cannot be made, however, in the case of Michigan. As noted above, Michigan repaid only the amount it had received from the Morris Canal and Banking Company, and it too seems to have regained access to credit markets fairly rapidly. At the time of the settlement, one commentator wrote “Michigan has passed a law ... cunningly devised to slip their neck out of the noose and obtain from their creditor an acquiescence in their repudiation” (quoted in McGrane, 1935 p. 167). By the mid-1850's, yields on Michigan bonds were under 6 percent (U.S. Department of Interior, 1884 v. 7, p. 628).

Yields on the bonds of states that defaulted temporarily in the early 1840's, but eventually paid their debts in full, remained elevated for several years after their resumptions. The spreads of these yields over those on Treasury bonds declined over time, however, presumably as a result of lower perceived risk.⁸ For example, Pennsylvania resumed payments in 1845, but yields on state bonds remained about 4 percentage points higher than those on Treasuries for several years. By the early 1850's, however, yields on Pennsylvania bonds were only 1–2 percentage points higher than those on Treasuries (see Figure 1). Similarly, yields on Maryland bonds were close to those on Treasury bonds by the early 1850's (*Hunt's Merchant Magazine and Commercial Review*, 1853 v. 28(4), p. 488).

As noted above, Indiana and Illinois both received new loans to allow the completion of their canals. These loans presumably did not reflect an improvement in the states' reputations for repayment since they preceded the states' resumptions. Rather the creditors likely believed that without the completion of the canals, the old loans would not be repaid, but with their completion, the loans might be

⁸ Similarly, Sule Ozler (1993 pp. 610–12) notes that more recent defaults had a larger effect on the interest rates charged on developing country debts in 1968–1981 than did earlier defaults.

served. In addition, the states transferred ownership of the canals to the borrowers, making subsequent repudiation of their claims less likely.⁹ Nonetheless, steady payment on the states' debts appears to have improved their reputations by the 1850's. Yields on Indiana bonds, which peaked at nearly 32 percent in 1842, fell to within 2 percentage points of Treasury yields by the early 1850's (Figure 1). Similarly, yields on Illinois bonds, which were even higher than those on Indiana bonds in the early 1840's, declined to less than 6 percent by 1855 (James, 1938 v. 1, pp. 143, 172).

Most of the states that settled their defaults in the 1840's increased their borrowing little in the period before the Civil War. As a result, they generally paid out in debt service more than they received in "new money." This fact should not be taken as evidence against the view that the states repaid in order to maintain access to capital markets. Rather it may be that the situations under which the states would have borrowed more did not arise. Indeed, reputation models generally imply the existence of lending ceilings (as noted in Eaton and Gersovitz [1981]), and some states may have been near their ceilings, making additional borrowing unlikely. Nonetheless, states near their borrowing ceilings may choose to service their debts because they value the ability to borrow again following a period of net repayment.

C. Whose Reputation?

If reputational effects sustain sovereign debt markets, then it is important to know exactly whose reputation matters. In the case of a democracy, it seems likely that the reputation is that of the electorate. If this is so, then the defaults and repudiations of some states could affect the reputation of the entire country because the repudiating states' citizens are also citizens of the country.

⁹ But perhaps not impossible. The Fifth Amendment requires the federal government to provide compensation for property taken under eminent domain, but it does not require states to do so. The Fourteenth Amendment, which does restrict states' ability to seize property was not passed until after the Civil War. See the discussion in *American Jurisprudence*, 1974 v. 26, pp. 645–46.

Indeed, McGrane suggests that the reputation of the federal government in European capital markets was temporarily affected by the states' defaults. In 1842 the federal government investigated the possibility of selling bonds in Europe. The U.S. agents were told, however, that the U.S. bonds could not be sold there because European investors feared that the federal government would also default (McGrane, 1935 p. 33–34). It is possible, however, that the inability to sell federal bonds was the result of efforts by European banks to induce the federal government to assume the state debts. William Robinson, the commissioner sent to Europe to negotiate the loan, argued that the lenders had formed a cartel. As evidence he noted that the European bankers seemed willing to market a large loan if part of the funds were used to pay the debts of Mississippi, Michigan, Arkansas, Illinois, and Indiana. (See U.S. Congress, House of Representatives [1843a Document No. 197, p. 4]. English [1991] provides further discussion.)

Institutional changes also may affect a sovereign borrower's reputation. For example, Florida and Mississippi were able to issue bonds after the Civil War without reaching any settlement on their pre-war debts. On the one hand, these resumptions show that the old creditors could not block the states from borrowing—indicating that their lack of borrowing before the war was the result of their poor reputations rather than legal restrictions. On the other hand, they also suggest that observable changes in government—i.e., the imposition of northern-backed governments during Reconstruction—could affect reputation. In addition, foreign investors may have thought that the Union victory in the Civil War strengthened federal control of the states and would allow them to use the federal judiciary to force states to pay their debts.¹⁰

¹⁰ Orth (1987 Chapter 4) suggests that they were right. It was the political maneuvering following the election of 1876, he argues, that restored sovereign immunity to the states. Ironically, many southern states defaulted again when Reconstruction ended (see William A. Scott, 1974).

IV. Rebuilding Reputational Models of Sovereign Debt

The historical evidence presented here in favor of reputational models of sovereign debt contradicts the theoretical work in Bulow and Rogoff (1989a). This contradiction raises an obvious question: how could reputation have been sufficient incentive to repay debts?

One way to rehabilitate reputational models is to build models in which states have more than one market that depends on their reputation, as in Cole and Patrick Kehoe (1994).¹¹ Their paper provides a model in which sovereign borrowers repay because not doing so has large costs in another market in which the government's reputation for honesty matters. These sorts of "reputational spillovers" seem too small in this case, however. As noted above, Louisiana defaulted on its bank bonds, yet by paying its state debt proper it managed to recover its reputation in the bond market. Similarly, Michigan regained access to credit markets fairly quickly after its partial repudiation. Thus the reputational spillover from a portion of a state's debts to the rest was fairly small, suggesting that the spillover to other activities of the state would also be small.

The most obvious criticism of Bulow and Rogoff's argument is that actuarially fair payment-in-advance insurance contracts were not available in the 1840's. Without such contracts, reputational effects may be sufficient to enforce repayments. Even without them, however, asset markets may still be sufficiently rich to make defaulting desirable. This possibility seems unlikely in this case. The main shocks to states' demand for foreign financing were likely to come from improvements in transportation technology and wars—and there were no contracts available that were contingent on these events.

Furthermore, if a government tried to self-insure after a default by accumulating assets, it would have had to hold a substantial surplus

in some periods. This strategy, however, would have been difficult politically. As noted by Curtis (1844 p. 151), "The ... political parties, which have ruled this country since the adoption of the Constitution, agree, that any accumulation of money raised by taxation is not to be thought of, and that no more is to be drawn from the people than is absolutely necessary." Curtis goes on to argue that a state's reputation in credit markets is important specifically because U.S. states could not accumulate surpluses, and in an emergency they might need more resources than they could tax in a single year.¹²

The reason for the inability of states to accumulate surpluses is not given by Curtis, but it presumably was the result of U.S. citizens' distrust of government. A lack of trust in their government could cause citizens to choose to repay debts rather than repudiate them. If, for example, citizens believe that state officials will either steal or waste a large enough share of any accumulated surplus, then they will be willing to give up the gain to be had by repudiation in order to avoid having the state hold surpluses.

Certainly the political battle over the federal surplus in the mid-1830's was marked by claims that the Federal government would invest the money inefficiently if it were not distributed. In addition, some argued that the surplus was being used by the executive branch to enhance its powers at the expense of Congress (Edward G. Bourne, 1885 pp. 24–25). The decision to distribute the surplus to the states in 1837 suggests that these arguments were persuasive. In addition, many states distributed their shares of the surplus to local governments, generally to be used for education, suggesting distrust of state government. Indeed, in Maine the surplus was distributed directly to the population (Bourne, pp. 122–23).

¹¹ Indeed, this possibility is noted explicitly by Bulow and Rogoff (1989a p. 47). They point out that the cost of default in such a case is simply the cost of losing a good reputation in the other market; there is no additional cost to losing a good reputation in the market for loans.

¹² A similar point was made by Alexander Hamilton in the debate over the settlement of the Revolutionary War debt. See Peter M. Garber (1991) for a discussion of that case.

Note that there is no problem if Ricardian equivalence holds. However, states were borrowing in Europe specifically because Ricardian equivalence did not hold: private investors did not have access to European credit markets.

It is important to note that the federal surplus that was distributed in 1837 was relatively small compared to the surpluses that the states would have accumulated under the default strategy. The aggregate debt of the states in 1841 was over \$200 million—more than seven times the federal surplus that proved politically insupportable. Moreover, heavily indebted states had debts between 10 and 40 times as large as their share of the federal surplus.

V. Concluding Remarks

The defaults and repudiations of U.S. states in the 1840's are a powerful experiment for those studying sovereign debt. Under the United States Constitution states are sovereign and cannot be compelled to repay debts. Sanctions that could be imposed on other sovereign debtors, such as trade sanctions or military action, were very difficult to impose on U.S. states because of the ease of trade between states and the military power of the United States. In spite of the theoretical result in Bulow and Rogoff (1989a), reputation effects appear to have been sufficient to induce most states to repay. Evidently, some mixture of incomplete markets and political constraints made it impossible for state governments to implement the strategy suggested by Bulow and Rogoff.

Of course, the conclusion that states repaid their debts in order to maintain their reputations in debt markets leaves open the question of exactly how reputation was gained and lost at that time. There are a vast number of possible reputational models and equilibria to choose from, some of which put very few constraints on the pattern of repayment (see, e.g., Herschel I. Grossman and John B. Van Huyck, 1988). Indeed, repeated games generally suffer from a multiplicity of equilibria. Even debt models based on sanctions can have many equilibria if they are repeated games (Eaton and Maxim Engers, 1992 p. 905). Evidently, one must select the model and equilibrium that seems most reasonable given a particular historical situation. The experiences of defaulting U.S. states in the 1840's and 1850's are consistent with a signaling model like that in Cole et al. (1995). In that model, sovereign borrow-

ers are one of two types: myopic and nonmyopic. Myopic borrowers default on loans, while nonmyopic borrowers do not. A borrower's type changes over time. Cole et al. show that a reasonable restriction on the model yields equilibria in which a borrower that defaulted in the past but is now nonmyopic signals its change of type by making payments on its outstanding debts. In these equilibria, a settlement restores the borrower's reputation and gives it access to capital markets. These equilibria are appealing because "reputation" is not just a way to coordinate punishment, but actually summarizes lenders' information about the type of the borrower.

In any case, that the cost of default was a reputation-based loss of access to future loans appears to have been clear to observers at the time. In 1846 the Barings were quoted in *Niles Register* as believing that Europeans would not "purchase either old or fresh [state] securities until ... those states which were still defaulters had shown their willingness and ability to recommence and continue the regular payment of future dividends" (quoted in McGrane, 1935 p. 268). Similarly, the *London Times* wrote in that year that the defaulting states would eventually choose to pay their debts because they "will deem it a not disadvantageous transaction to lay out ten or twenty millions ... in purchasing a restoration of their forfeited respectability" (December 3, 1846, quoted in McGrane, p. 166).

DATA APPENDIX

A. Debt Data

The primary source for the debt data in Tables 1 and 2 is U.S. Congress, House of Representatives (1843b) Report No. 296, 27th Congress, 3rd Session. These data are supposed to be a complete statement of the States' debts as of September 2, 1841. State bonds are generally listed by the object of the borrowing, and the interest rate on the bonds is shown. These data allowed me to construct Table 1. I have included in all cases contingent debts. These were generally guarantees by the state to pay bonds issued by banks, railroads, etc. In some cases these bonds were included in the tables in the report, in others they were listed in footnotes. In some cases the interest rates on bonds were not reported. I have assumed that such bonds paid 6 percent—roughly the average rate. In some cases the bonds were listed in pounds sterling. I valued such bonds at \$4.86 to the pound. In some cases there were additional complications, which are discussed in the data appendix to English (1991).

B. State-Income Data

Richard A. Easterlin (1960) provides estimates of state income based on the 1840 census and the pioneering work of Seaman. These data are on a income-paid basis and so are something like net domestic (state) product. I took the per-capita levels of state income from Easterlin and multiplied by the ratio of national per-capita GNP in 1839 reported by Gallman (1964 p. 26) to Easterlin's national per-capita income figure. Gallman's national per-capita GNP number is also based on the 1840 census data. The result is an estimate of per-capita gross state products. These are shown in Table 2.

The state population data is taken from Easterlin (1960). In contrast to Arthur Grinath et al. (1995) I use the total population of the state rather than the white population.

C. Price Data

The Warren-Pearson Wholesale Price Index is taken from U.S. Department of Commerce (1975), Series E1, p. 115.

D. Bond Prices

The prices of the United States and Indiana state bonds before 1846 were taken from various issues of *Hunt's Merchants Magazine*. From 1846 through 1851 the data are from *Bankers' Magazine and Statistical Register*. I thank Warren Weber and Andrew Economopoulos for providing me with copies of this data. The Pennsylvania bond prices were taken from *Van Court's Counterfeit Detector and Bank Note List*. See Gary Gorton (1989) for a description of this periodical. I thank Gorton for making microfilmed copies of this journal available to me. I calculated the yields to maturity shown in Figure 1 based on reported bond prices and maturities. In some cases no maturity was shown; in such cases the yield is based on an estimate of the maturity.

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